

Lista 2 - Aplicações de funções polinomiais
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LINK do Doc com as fotos do caderno:

<https://docs.google.com/document/d/1q0nQp-af74-KLGrZ5kvI5nMUB6G7-lSgm4QseM-7blw/edit?usp=sharing>

PAG 1:

10]

y = cm

x = dias

1cm a cada dia

$$x = y \cdot 5$$

$$30 = y \cdot 5$$

$$\frac{30}{5} = t = 6$$

11]

a)

(1, 3) e (-1, -1)

$$y = ax + b$$

$$\{3 = a \cdot 1 + b\} s1$$

$$\{-1 = a(-1) + b\} s2$$

s2:

$$-1 = a(-1) + b \quad | \quad -1 = 2(-1) + b$$

$$-1 = -a + b \quad | \quad -1 = -2 + b$$

$$b = a - 1 \quad | \quad 1 = b$$

s1:

$$3 = a \cdot 1 + b$$

$$3 = a \cdot 1 + (a - 1)$$

$$3 = a \cdot 1 + a - 1$$

$$3 = 2a - 1$$

$$4 = 2a$$

$$\frac{4}{2} = a = 2$$

$$2$$

$$y = 2x + 1$$

b)

(1, 1) e (2, -1)

$$\{1 = a \cdot 1 + b\} s1$$

$$\{-2 = a \cdot 2 + b\} s2$$

s1:

$$1 = a \cdot 1 + b \quad | \quad 1 = a \cdot 1 + 4$$

$$1 = a + b \quad | \quad 1 - 4 = a$$

$$a = 1 - b \quad | \quad -3 = a$$

s2:

$$-2 = a \cdot 2 +$$

$$-2 = (1-b)2 + b$$

$$-2 = 2 - 2b + b$$

$$-4 = -b \quad (\cdot -1)$$

$$\mathbf{4 = b}$$

$$\mathbf{y = -3x + 4}$$

14]

a)

(0, 3000) e (8, 3500)

$$\{3000 = a \cdot 0 + b\} \quad \mathbf{3000 = b}$$

$$\{3600 = a \cdot 8 + b\}$$

$$3600 = a \cdot 8 + 3000$$

$$\mathbf{600 = a = 75}$$

$$\mathbf{8}$$

$$y = 75x + 3000$$

b)

$$6000 = 75x + 3000$$

$$6000 - 3000 = 75x$$

$$3000/75 = x = 40$$

$$\mathbf{1985 + 40 = 2025}$$

15]

a)

gráfico

b)

$$f(0) = 20 + 5 \cdot 0$$

$$y = 20 + 0$$

$$\mathbf{y = 20}$$

c)

$$f(180) = 20 + 3 \cdot 180$$

$$y = 20 + 540$$

$$\mathbf{y = 560}$$

d)

$$30 \text{ a } 50 \text{ pois } T(t) = 170$$

17]

a)

Cia do Corpo

$$f(t) = 45t + 90$$

Chega de Moleza

$$f(t) = 50t + 70$$

b)

Cia do Corpo : 630

$$y = 45.12 + 90$$

$$y = 540 + 90$$

$$y = 630$$

Chega de moleza: 670

$$y = 50.12 + 70$$

$$y = 600 + 70$$

$$y = 670$$

R: cia do corpo

20]

$$c = 2000 + 0,8x$$

$$c = 2000 + 0,8 \cdot 1000$$

$$c = 2000 + 800$$

$$\mathbf{c = 2800}$$

22]

a)

v = vendas

$$f(v) = 0,05v + 800$$

b)

$$5h = 500 = 200 \cdot 0,05 = 10 \text{ por hora}$$

$$s(x) = 10x + 800$$

$$\text{se } 220 = 10.220 + 800 = 3000$$

com aumento de 20%

$$10.220 + 800.1,2 = 3160$$

com 6%

$$\mathbf{200.0,06.220 + 800 = 3440}$$

PAG 2:

7]

$$F(f) = -t^2 + bt - 156$$

$$vm = -b/2.a$$

$$14 = -b/2.(-1)$$

$$14 = -b/-2$$

$$12(-2) = -b$$

$$-28 = -b (. -1)$$

$$28 = b$$

10]

80

p/l

$$80/4 = 20$$

12]

$$L = r - c$$

$$L = 6000x - x^2 - (x^2 - 2000x)$$

$$6000x - x^2 - x^2 - 2000x$$

$$8000x - 2x^2 (/2)$$

$$4000x - x^2$$

$$Xv = 8000/2.(-1) = 2000$$

18]

$$-2t+8t(t>0)$$

$$a) x = -b \pm \sqrt{\Delta} / 2.a$$

$$-(8) \pm \sqrt{64} / 2(-2)$$

$$-8 \pm 8 / -4$$

$$s1 +$$

$$-8 + 8 / -4$$

$$0 / -4 = x0$$

$$s2 -$$

$$-8 - 8 / -4$$

$$-16 / -4 = x4$$

|
|
|
|

$$0 --- 4$$

b)

$$Yv = -64 / 4(-2) = 8$$

22]

$$f(x) = 2(x-1)^2 - 4$$

$$(x-1)(x-1) = x^2 - 2x + 1$$

$$f(x) = 2(x^2 - 2x + 1) - 4$$

$$f(x) = 2x^2 - 4x + 2 - 4$$

$$f(x) = 2x^2 - 4x - 2$$

$$(x=-2 \text{ e } y=4)$$

$$f(x) = 2x^2 - 4x - 2$$

$$4 = 2(-2)^2 - 4(-2) - 2$$

$$4 = 8 + 8 - 2$$

$$4 = 14$$

$$(x=0 \text{ e } y=-2)$$

$$-2 = 2(0)^2 - 4(0) - 2$$

$$-2 = 0 - 0 - 2$$

$$\mathbf{-2 = -2}$$

R = alternativa de I e III

$$25]$$

$$(4, 12)$$

$$F(x) = ax^2 + bx$$

$$4^2a + 4b = 12$$

$$16a + 4b = 12 \quad (/4)$$

$$\mathbf{4a + b = 3}$$

$$(8, 0)$$

$$F(x) = ax^2 + bx$$

$$8^2a + 8b = 0$$

$$64a + 8b = 0 \quad (/8)$$

$$\mathbf{8a + b = 0}$$

Sistema:

$$4a + b = 3$$

$$8a + b = 0$$

para A

$$4a + b = 3 \quad (. \cdot -1)$$

$$8a + b = 0$$

$$-4a - b = -3$$

$$\underline{8a + b = 0}$$

$$4a = -3$$

$$a = -1/4$$

para B

$$4a + b = 3 \quad (. \cdot -2)$$

$$8a + b = 0$$

$$-8a - 2b = -6$$

$$\underline{8a + b = 0}$$

$$\underline{0 - b = -6}$$

$$b = -6/-1 = 6$$

R: letra C

27]

(3,9) e (6,0)

$$9 = a(3)^2 + b(3)$$

$$9 = 9a + 3b \quad (/3)$$

$$\mathbf{3 = 3a + b}$$

$$0 = a(6)^2 + b(6)$$

$$0 = 36a + 6b \quad (/6)$$

$$\mathbf{0 = 6a + b}$$

sistema

$$\mathbf{3 = 3a + b}$$

$$\mathbf{0 = 6a + b}$$

para A

$$3 = 3a + b \quad (-1)$$

$$0 = 6a + b$$

$$-3 = -3a - b$$

$$0 = 6a + b$$

$$-3 = 3a$$

$$-3/3 = a = -1$$

para B

$$3 = 3a + b \quad (-2)$$

$$0 = 6a + b$$

$$-6 = -6a - 2b$$

$$0 = 6a + b$$

$$-6 = -b \quad (-1) = 6$$

Letra D -1,6,0